

Method used	Dataset size	Testing-set predictive performance	Time taken for the model to be fit
XGBoost in Python via scikit-learn and 5-fold CV	100	0.85	0.266
	1000	0.945	0.404
	10000	0.972	1.631
	100000	0.984	6.325
	1000000	0.9875	48.109
	10000000	0.988	442.848
XGBoost in R – direct use of xgboost() with simple cross-validation	100	0.9474	2.05
	1000	0.9598	3.01
	10000	0.9750	5.53
	100000	0.9850	20.32
	1000000	0.9869	156.32
	10000000	0.9872	1135.44
XGBoost in R – via caret, with 5-fold CV simple cross-validation	100	0.94	1.98
	1000	0.96202	4.84
	10000	0.9831	25.96
	100000	0.9901	241.957

Method used	Dataset size	Testing-set predictive performance	Time taken for the model to be fit
	1000000	0.9906	985.263
	10000000	0.9845	1985.765

The XGBoost implementation through caret in R with 5-fold cross-validation stands as my recommended approach based on the analysis of Python and R platforms. The data demonstrates that this recommendation produces the best predictive results because it maintains high accuracy levels at 0.9906 when working with 1,000,000 data points. The R-caret implementation delivers better performance in the essential middle range of data volume which covers most practical application areas even though the Python implementation matches its accuracy levels.

The time-performance analysis demonstrates important considerations for recommending this approach. The Python implementation delivers faster performance during smaller data processing tasks by completing 100,000 data points within 6.325 seconds whereas R-caret requires 241.957 seconds. The R-caret approach provides superior predictive accuracy than the other methods (0.9901 versus 0.984) possibly making its computational cost acceptable when prediction quality takes priority. The R-caret method demonstrates performance deterioration to 0.9845 when processing the largest dataset of 10,000,000 data points which indicates potential challenges in optimizing extreme scale operations.

The R implementation without caret provides a practical solution for organizations needing different modeling capabilities because it delivers 0.9869 predictive accuracy alongside 156.32 seconds of training time for 1,000,000 data points. The method provides organizations with a

useful compromise solution between prediction precision and processing speed. The R-caret implementation stands as the optimal choice for practical applications because it delivers superior results when dealing with typical data ranges from thousands to millions of records but requires an alternative solution only in cases of extreme time pressure or large data volumes.